

LD9 Master Theory A to Z

Final Documentation Package for the Bank Marketing Subscription Prediction Project

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This document explains the full theory behind the ensemble learning project in a simple but complete way. It is written to support learning, revision, and interview preparation.

1. What is Ensemble Learning?

- Ensemble learning means combining multiple models instead of relying on one single model.
- The main idea is that several models can work together and create a stronger final prediction.
- Different ensemble methods reduce different kinds of model weakness. Some reduce variance, some reduce bias, and some learn how to combine model strengths.

2. Weak Learners and Strong Learners

- A weak learner is a model that is only moderately good by itself.
- A strong learner is a model that performs much better and generalizes well.
- Ensemble methods often turn many moderate learners into one stronger system.

3. Bagging

- Bagging stands for Bootstrap Aggregating.
- It creates many bootstrap samples by drawing rows from the training data with replacement.
- A separate model is trained on each sample, and the predictions are combined by voting or averaging.
- Bagging mainly reduces variance and makes predictions more stable.

4. Bootstrap Samples

- A bootstrap sample is built by randomly sampling training rows with replacement.
- With replacement means the same row can appear multiple times, while some rows may be left out.
- These left-out rows are useful for Out-of-Bag validation.

5. Out-of-Bag (OOB) Score

- In Random Forest, each tree does not see every training row.
- The rows not used by a tree are called out-of-bag rows for that tree.
- By collecting predictions on these unseen rows, we get an internal validation score without creating a separate validation split.

6. Random Subspace Method (RSM)

- RSM means each tree looks at only a random subset of features when making a split.
- This makes trees less similar to each other.
- Less correlation between trees helps the forest become stronger as an ensemble.

7. Random Forest

- Random Forest combines bagging, decision trees, random feature selection, and voting.
- Each tree sees different rows and different feature subsets.
- The forest prediction is the combined decision of all trees.
- Random Forest usually performs better and more stably than one unpruned decision tree.

8. Feature Importance

- Tree-based models can estimate which features contributed most to prediction decisions.
- This helps us understand which customer attributes matter more for the subscription outcome.
- Feature importance should be interpreted carefully because it is not the same as causal impact.

9. Cross-Validation (CV)

- Cross-validation splits the training data into several folds.
- In 5-fold CV, the model trains on 4 folds and validates on the 1 remaining fold, repeating this process 5 times.
- The final CV score is the average across folds. This gives a more stable estimate than one single split.

10. Stacking

- Stacking combines different model types instead of many copies of the same model type.
- The first-level models are called base learners.
- Their predictions are used as new input features for a meta-model.
- The meta-model learns which base models to trust more in different situations.

11. Model Diversity

- Stacking works best when the base learners are diverse.
- Diversity means the models do not make exactly the same mistakes.
- For example, a Random Forest, Logistic Regression, KNN, and Decision Tree have different learning logic, so combining them can be helpful.

12. OOF Predictions

- OOF stands for Out-of-Fold.
- In manual stacking, each training row must be predicted by a model that did not train on that row.

- These fair predictions are called OOF predictions and they become the training features for the meta-model.
- This prevents data leakage.

13. Blending

- Blending is a simpler version of stacking.
- Instead of using multiple CV folds, it uses one split: one part trains base models and another part trains the meta-model.
- Blending is easier to implement but depends more on a single split and uses less data for base-model training.

14. Three-Layer Stacking

- Three-layer stacking adds an extra stacking layer.
- Level 0 contains base learners.
- Level 1 contains intermediate meta-models trained on Level-0 predictions.
- Level 2 is the final model trained on Level-1 predictions.
- This can work, but extra complexity does not always bring enough additional performance.

15. Boosting

- Boosting builds models sequentially instead of in parallel.
- Each new model focuses on correcting mistakes made by previous models.
- Boosting tries to reduce bias and create a stronger final model.

16. Gradient Boosting

- Gradient Boosting is a boosting method where each new tree learns from the residual errors of the previous stage.
- It is powerful and flexible, but it can be more sensitive to parameter choices than Random Forest.

17. Evaluation Metrics

- Accuracy = overall share of correct predictions.
- Precision = among predicted subscribers, how many were truly subscribers.
- Recall = among true subscribers, how many the model correctly found.
- F1-score = balance between precision and recall.
- ROC-AUC = how well the model ranks positive cases above negative cases.

18. Final Theory Lesson from the Project

- A more complex model is not always the best practical model.

- The best model is the one that gives a strong balance of performance, stability, interpretability, and maintainability.
- In this project, the RF-meta stacker was the best practical compromise.